

Grain Dryer Simulator

Real-time drying control panel — feature overview

A simulated PLC-style dashboard for drying agricultural crops (Soyabean, Tur, Chili).

This document summarizes the capabilities of the interactive grain-dryer simulation. The app models heat, airflow, moisture loss and load-cell weight in real time, and presents them through an industrial control panel with live telemetry, trend charts and batch reporting.

LIVE MONITORING & TELEMETRY

- **Sensor panel**
Live readouts for chamber temperature, relative humidity, grain moisture and airflow, updated continuously as the batch dries.
- **Load-cell reading**
Real-time grain weight from the simulated load cell, shown on the panel so you can watch mass drop as water evaporates.
- **Chamber schematic**
A visual diagram of the drying chamber showing heater, fan and airflow state at a glance.
- **Status header**
Prominent mode indicator (Standby, Drying, Complete, Fault, E-Stop) plus key setpoints.

SIMULATION & CONTROL

- **Crop presets**
One-click presets for Soyabean, Tur and Chili, each with recommended target moisture and drying temperature.
- **Adjustable setpoints**
Manually tune target moisture and temperature to run custom drying profiles.
- **Auto-stop at target**
The heater automatically switches off and the dryer enters cool-down the moment the crop reaches its target moisture.
- **Emergency stop**
An E-Stop control immediately halts the process, with a guarded reset before resuming.

INSIGHTS & REPORTING

- **Estimated time remaining**
A live ETA computed from the current drying rate and the remaining moisture gap, alongside elapsed run time.

- **Energy & cost estimate**

Running energy consumption (kWh) and operating cost, calculated from runtime and heater temperature.

- **Drying trend charts**

Time-series charts plotting moisture, temperature, humidity and weight across the batch.

- **Quality warnings**

Alerts for over-temperature and stalled drying so grain quality is protected.

- **Batch history log**

Every completed cycle is recorded for comparison — crop, duration, moisture in/out, water removed, average rate, peak temperature, energy and cost — with running totals and a clear option.

- **Batch summary report**

An end-of-run report card summarizing the completed drying cycle.

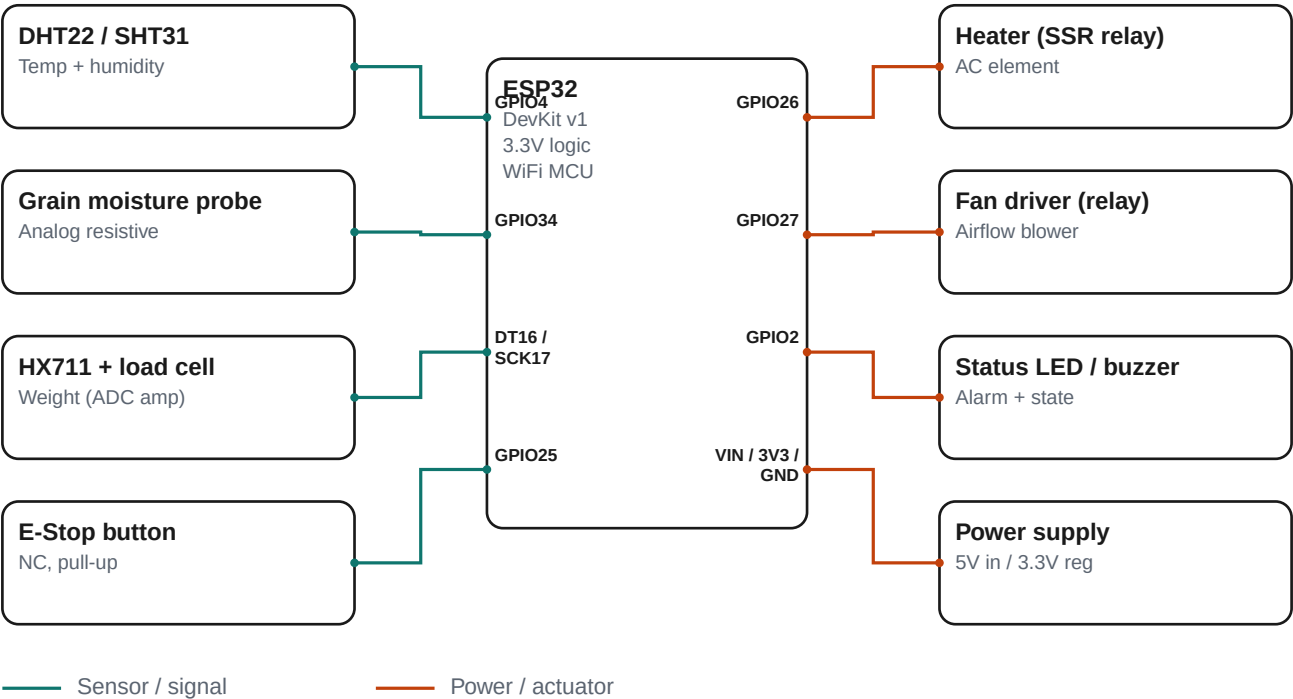
DIAGNOSTICS

- **Event log**

A timestamped stream of system events (info, warnings and faults) for tracing exactly what happened during a run.

WIRING / CIRCUIT DIAGRAM (ESP32)

Hardware wiring that maps the simulation to real components. An ESP32 reads the temperature/humidity, grain-moisture and load-cell sensors, monitors the emergency-stop input, and drives the heater and fan through relays.



Component	ESP32 pin	Notes
DHT22 / SHT31 sensor	GPIO4	Chamber temperature & humidity (3.3V, 4.7k pull-up on data)
Grain moisture probe	GPIO34 (ADC)	Analog input; use voltage divider, ADC1 channel
HX711 load-cell amp	GPIO16 / GPIO17	DT = 16, SCK = 17; measures grain weight
Emergency-stop button	GPIO25	Normally-closed, internal pull-up, triggers E-Stop
Heater SSR relay	GPIO26	Drives solid-state relay for the heating element
Fan / blower relay	GPIO27	Controls airflow blower (PWM capable)
Status LED / buzzer	GPIO2	Mode indicator and fault alarm
Power supply	VIN / 3V3 / GND	5V input, on-board 3.3V regulator, common ground

Note: mains-voltage wiring to the heater and fan must go through appropriately rated relays/SSRs and isolation. Pin assignments are a reference and can be remapped in firmware.

